Day 3 Study Notes

Sun 15 Nov 2026 · Sprint 60 · JEE Main 2027

Part	Time	What to do
Study	3 h 30 min	Physics: Measuring instruments and experimental skills (75 min) · Maths: 3D geometry (part 2) (75 min) · Chemistry: Coordination compounds (part 2: isomerism) (60 min)
Exam	1 h 15 min	The Day 3 exam in the Sprint 60 app: 30 questions, timed, no notes.
Review	1 h 15 min	Open every wrong and skipped question in the app, read its solution, tag the reason, then use Retry my mistakes.

How to use each chapter note. Read it once without stopping. Cover the formulas and write them from memory. Work the solved examples on paper before reading the working. Then read the NCERT section listed at the end.

Physics · Measuring instruments and experimental skills

Units and measurements was one of the two most-asked Physics chapters of 2026, and a large share of those questions were vernier and screw gauge readings. Every one is solved with the same three-line method.

1. Least count

The least count is the smallest length an instrument can measure. Every reading is (main scale reading) + (number on the fine scale) × (least count).

2. Vernier calipers

- If n vernier scale divisions (VSD) equal $n-1$ main scale divisions (MSD): $\mathrm{LC} = 1\,\mathrm{MSD} - 1\,\mathrm{VSD} = \dfrac{1\,\mathrm{MSD}}{n}$.
- General case, n VSD $=$ m MSD: $\mathrm{LC} = \left(1 - \dfrac{m}{n}\right) \times 1\,\mathrm{MSD}$.
- Standard instrument: 1 MSD $= 1$ mm and 10 VSD $= 9$ MSD, so LC $= 0.1$ mm $= 0.01$ cm.
- Reading $=$ main scale reading $+$ (coinciding vernier division) $\times$ LC.

Zero error is the reading when the jaws are closed.

Type	What you see with jaws closed	Zero error
Positive	vernier zero is to the right of the main scale zero	$+(\text{coinciding division}) \times \mathrm{LC}$
Negative	vernier zero is to the left of the main scale zero	$-(n - \text{coinciding division}) \times \mathrm{LC}$

Corrected reading = observed reading − zero error, keeping the sign of the zero error. A positive error is subtracted; a negative error ends up being added.

> **Solved example 1.** LC $= 0.01$ cm. With the jaws closed, the vernier zero lies to the right of the main zero and the 4th division coincides. An object reads 3.1 cm on the main scale with the 8th division coinciding. Find its length.
> Zero error $= +4 \times 0.01 = +0.04$ cm. Observed reading $= 3.1 + 8 \times 0.01 = 3.18$ cm. Corrected length $= 3.18 - 0.04 = 3.14$ cm.

3. Screw gauge

- Pitch $=$ distance the screw advances in one full rotation.
- $\mathrm{LC} = \dfrac{\text{pitch}}{\text{number of circular scale divisions}}$. Common values: pitch 1 mm with 100 divisions, or 0.5 mm with 50 divisions; both give 0.01 mm.
- Reading $=$ linear scale reading $+$ (circular scale reading) $\times$ LC.

Type	With the studs touching	Zero error
Positive	circular zero is below the reference line; a small division such as 4 coincides	$+4 \times \mathrm{LC}$
Negative	circular zero is above the reference line; a large division such as 97 coincides	$-(100 - 97) \times \mathrm{LC}$

The correction rule is the same: corrected $=$ observed $-$ zero error. Backlash error is avoided by turning the screw in one direction only.

Solved example 2. Pitch 1 mm, 100 divisions. Zero error is -0.03 mm. A wire reads 3 mm on the linear scale and 46 on the circular scale.
Observed $= 3 + 46 \times 0.01 = 3.46$ mm. Corrected $= 3.46 - (-0.03) = 3.49$ mm.

4. Simple pendulum experiment

- $T = 2\pi\sqrt{\dfrac{L}{g}}$, so $g = \dfrac{4\pi^2 L}{T^2}$. L is the string length plus the radius of the bob.
- Time n oscillations (usually 20 to 50) and divide: $T = \dfrac{t}{n}$. This divides the stopwatch error by n.
- Percentage error: $\dfrac{\Delta g}{g} = \dfrac{\Delta L}{L} + 2\dfrac{\Delta T}{T}$, and $\dfrac{\Delta T}{T} = \dfrac{\Delta t}{t}$.
- The graph of L against T^2 is a straight line through the origin with slope $\dfrac{g}{4\pi^2}$.

5. Resonance tube (speed of sound)

- First resonance: $l_1 + e = \dfrac{\lambda}{4}$. Second resonance: $l_2 + e = \dfrac{3\lambda}{4}$.
- Subtracting removes the end correction: $\lambda = 2(l_2 - l_1)$ and $v = 2f(l_2 - l_1)$.
- End correction: $e = \dfrac{l_2 - 3l_1}{2} \approx 0.3\,d$, where d is the inner diameter of the tube.

Traps. Check the units of the answer options: least counts are given in mm, cm and sometimes micrometres. When the question gives "n VSD coincide with m MSD" and 1 MSD is not 1 mm, compute the least count from scratch. A negative zero error makes the corrected reading larger than the observed one. In the pendulum, the error in T is doubled because of T^2.

Read in NCERT: Physics Class 11 Part 1, chapter "Units and Measurements", and the vernier calipers, screw gauge, simple pendulum and resonance tube experiments in the Class 11 laboratory manual.

Mathematics · 3D geometry (part 2)

Shortest distance, foot of perpendicular and image are the three most-asked 3D results. All of them come from one idea: write a general point on the line and apply a perpendicularity condition.

1. Shortest distance between two skew lines

Skew lines are neither parallel nor intersecting. For $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu\vec{b}_2$:

$$d = \frac{\left|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)\right|}{\left|\vec{b}_1 \times \vec{b}_2\right|}$$

- The lines intersect (are coplanar) when $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0$ and they are not parallel.
- In Cartesian form the numerator is the determinant with rows $(x_2 - x_1,\ y_2 - y_1,\ z_2 - z_1)$, (a_1, b_1, c_1) and (a_2, b_2, c_2).

> **Solved example 1.** Find the shortest distance between $\dfrac{x-1}{2} = \dfrac{y-2}{3} = \dfrac{z-3}{4}$ and $\dfrac{x-2}{3} = \dfrac{y-4}{4} = \dfrac{z-5}{5}$.
> $\vec{b}_1 \times \vec{b}_2 = (3\cdot 5 - 4\cdot 4,\ 4\cdot 3 - 2\cdot 5,\ 2\cdot 4 - 3\cdot 3) = (-1, 2, -1)$, with magnitude $\sqrt{6}$. $\vec{a}_2 - \vec{a}_1 = (1, 2, 2)$. Dot product $= -1 + 4 - 2 = 1$. So $d = \dfrac{1}{\sqrt{6}}$.

2. Distance between parallel lines, and from a point to a line

- Parallel lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}$ and $\vec{r} = \vec{a}_2 + \mu\vec{b}$: $d = \dfrac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|}$.
- Point P to the line through A with direction $\vec{b}$: $d = \dfrac{|\vec{AP} \times \vec{b}|}{|\vec{b}|}$.

3. Foot of the perpendicular and image

1. Write a general point $Q = (x_1 + a\lambda,\ y_1 + b\lambda,\ z_1 + c\lambda)$ on the line.
2. $\vec{PQ}$ must be perpendicular to the line: $\vec{PQ} \cdot (a, b, c) = 0$. Solve for λ. Shortcut: $\lambda = \dfrac{\vec{AP} \cdot \vec{b}}{|\vec{b}|^2}$.
3. Substitute λ to get the foot Q. The perpendicular distance is $|PQ|$.
4. The image P' of P in the line has Q as the midpoint of PP': $P' = 2Q - P$.

> **Solved example 2.** Find the foot of the perpendicular from $P(1, 2, 3)$ to $\dfrac{x-6}{3} = \dfrac{y-7}{2} = \dfrac{z-7}{-2}$, the distance, and the image of P.
> $Q = (6 + 3\lambda,\ 7 + 2\lambda,\ 7 - 2\lambda)$ and $\vec{PQ} = (5 + 3\lambda,\ 5 + 2\lambda,\ 4 - 2\lambda)$. Perpendicularity: $3(5 + 3\lambda) + 2(5 + 2\lambda) - 2(4 - 2\lambda) = 17 + 17\lambda = 0$, so $\lambda = -1$. Foot $Q = (3, 5, 9)$. Distance $= \sqrt{4 + 9 + 36} = 7$. Image $= 2Q - P = (5, 8, 15)$.

4. Point of intersection of two lines

Write general points with parameters λ and μ, equate two coordinates to find λ and μ, and check the third. If the third does not agree, the lines do not meet. Questions asking for an unknown constant so that two lines intersect are solved this way.

Traps. Before using any formula, check whether the lines are parallel: for parallel lines $\vec{b}_1 \times \vec{b}_2 = \vec{0}$ and the skew formula breaks down. Put each line in standard form first, with coefficients of x, y, z equal to $+1$. In the image, double the foot and subtract the point; do not double the point.

Read in NCERT: Mathematics Class 12 Part 2, chapter "Three Dimensional Geometry": shortest distance between two lines, with the solved examples and the miscellaneous exercise.

Chemistry · Coordination compounds (part 2: isomerism)

Isomer counting is the most common single question type from coordination compounds. Learn the table of standard cases in section 3; most questions are one of them.

1. Structural isomerism

Type	What differs	Example
Ionisation	an ion swaps places between the coordination sphere and the counter ion	$[Co(NH_3)_5SO_4]Br$ and $[Co(NH_3)_5Br]SO_4$
Hydrate (solvate)	water inside the sphere or outside as water of crystallisation	$[Cr(H_2O)_6]Cl_3$ (violet) and $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$ (grey-green)
Linkage	an ambidentate ligand bonds through a different atom	$[Co(NH_3)_5(NO_2)]Cl_2$ (yellow, through N) and $[Co(NH_3)_5(ONO)]Cl_2$ (red, through O)
Coordination	ligands swap between the cationic and anionic complex ions	$[Co(NH_3)_6][Cr(CN)_6]$ and $[Cr(NH_3)_6][Co(CN)_6]$

Ionisation isomers are told apart by tests on the solution: the first one above gives a precipitate with $AgNO_3$ (free Br^-), the second with $BaCl_2$ (free SO_4^{2-}).

2. Stereoisomerism

Geometrical isomerism arises when ligands can be next to each other (cis) or opposite (trans). **Optical isomerism** arises when a complex and its mirror image cannot be superimposed; such a complex has no plane of symmetry.

- Tetrahedral complexes do not show geometrical isomerism, because all four positions are next to one another.
- Square planar complexes show geometrical isomerism but, as a rule, not optical isomerism.
- Octahedral complexes show both. Optical isomerism in the syllabus always involves didentate ligands.

3. Standard cases to know by heart

Formula type	Shape	Example	Isomers
MA_2B_2	square planar	$[Pt(NH_3)_2Cl_2]$	2 geometrical (cis, trans)
MABCD	square planar	$[Pt(NH_3)(Br)(Cl)(py)]$	3 geometrical
MA_4B_2	octahedral	$[Co(NH_3)_4Cl_2]^+$	2 geometrical (cis, trans)
MA_3B_3	octahedral	$[Co(NH_3)_3Cl_3]$	2 geometrical (fac, mer)
MA_5B, MA_6	octahedral	$[Co(NH_3)_5Cl]^{2+}$	none
$M(AA)_3$	octahedral	$[Co(en)_3]^{3+}$, $[Cr(ox)_3]^{3-}$	2 optical (a mirror-image pair)
$M(AA)_2B_2$	octahedral	$[CoCl_2(en)_2]^+$	3 in total: trans (inactive) and a cis mirror-image pair
$M(AA)B_2C_2$	octahedral	$[Co(NH_3)_2Cl_2(en)]^+$	4 in total: two inactive forms and one mirror-image pair

fac and mer. In MA_3B_3, the facial (fac) isomer has the three identical ligands on one face of the octahedron, all next to each other. The meridional (mer) isomer has them around a meridian, with two of them opposite.

> **Why cis-$[CoCl_2(en)_2]^+$ is optically active but the trans form is not.** In the trans form the two Cl ligands are opposite, and the plane containing both en rings is a plane of symmetry. In the cis form no such plane exists, so the complex and its mirror image are different.

Method for counting. First decide the shape from the coordination number and metal (Pt^{2+}, Pd^{2+} and $[Ni(CN)_4]^{2-}$ are square planar). Match the formula to a row of the table. If the question asks for stereoisomers, add the optical isomers to the geometrical ones; if it asks for geometrical isomers only, count a mirror-image pair as one.

Traps. "Total number of isomers" and "number of geometrical isomers" are different questions. Linkage isomerism needs an ambidentate ligand: NO_2^-, SCN^- or CN^-. Coordination isomerism needs both the cation and the anion to be complex ions. A complex with a plane of symmetry is never optically active.

Read in NCERT: Chemistry Class 12 Part 1, chapter "Coordination Compounds", the section on isomerism, with its diagrams and in-text questions.